

PRARABDHA SONI

Senior Software Engineer(SDE-3)

prarabdha21@gmail.com | +91-8118898113 | www.linkedin.com/in/prarabdha-soni

Education

B.TECH CSE (Computer Science and Engineering) - IIT Guwahati

Sep'21

Senior Software Engineer

With over 4 years of experience in advanced backend technologies including Spring Boot, Node.js, Apache Kafka, Docker, Redis, FastAPI, python, GenAI, TensorFlow and AWS services like Cognito etc .

- Led a backend team of 5 as part of the operational systems pod, owning architecture and delivery of mission-critical lending workflows (disbursement, mandates, repayments)
- Drove end-to-end integration of third-party systems (BBPS, E-NACH, eSign), improving transaction reliability and enabling scalable loan lifecycle automation

Work Experience (4+ yrs exp.)

1) **Project Name: SaaS Product**

Company Name: Arthmate(Omnifi)

Core Infrastructure

Oct'23 –Feb'24

Backend, team size-8

Java, Spring Boot, Kafka, Redis, AWS, (Gateway, Cognito, etc.), PostgreSQL, MySQL MongoDB, Swagger

- **Automated Task Manager:** Architected an advanced task manager leveraging recursive Apache Kafka event processing. Fully configurable via a Master Data Management service, automating mission-critical workflows such as **loan disbursement**.
- Designed a dynamic runtime execution framework enabling condition-based service resolution and extensible workflows across loan processing pipelines

Credit Risk Modeling

Feb'24 – July'24

Risk team size – 7

Python, XGBoost, Scikit-Learn, FastAPI, AWS Lambda, Graph Neural Networks (GNNs), TensorFlow, LlamaIndex

- Developed a real-time Credit Risk Model using **XGBoost & LlamaIndex**, integrating CIBIL, CRIF, and Experian credit scores for loan default prediction and fraud detection.
- Engineered feature pipelines for borrower risk assessment, including credit history, debt-to-income ratio (**DTI**), loan stacking behavior, and transactional anomalies.
- Built a Credit Decisioning API with **FastAPI** and AWS **Lambda**, reducing loan approval time by 60% while ensuring compliance with RBI guidelines.
- Designed an AI-driven Fraud Detection System leveraging Graph Neural Networks (**PyTorch Geometric, DGL**) to analyze borrower relationships, flagging synthetic identity fraud and collusion networks.

2) **Project Name: LMS/LOS Origin Product**

Company Name: Arthmate

Core Infrastructure

Feb'22 – Present

Backend, team size – 15

Java, Spring Boot, Springbatch, Node.js,

Express.js, MongoDB, MySQL, Jest, Mocha, Docker, BullMQ, Redis, Passport.js

- Led **migration** of EC2-hosted Spring scheduled jobs to AWS **ECS Fargate**, redesigning the system from cron-based execution to a demand-driven, event-triggered architecture; improved scalability, reduced idle compute costs, and enhanced fault isolation and operational reliability.
- **Architected MSME** financial workflow validation system by introducing a centralized validator microservice integrated via Bull queue; enforced strict deviation-first processing to ensure completion of all compliance checks (including Udyam verification) before loan progression.
- **Designed and implemented BBPS integration** using serverless architecture (AWS Lambda, Cognito), enabling secure, scalable bill payment processing; built real-time EMI computation during transactions and supported multi-bank routing for apps like PhonePe.
- **Built and scaled** secure banking integrations with Axis, ICICI, and RBL, along with middleware connectivity to credit bureaus (CIBIL, CRIF); strengthened underwriting pipelines and improved loan disbursement efficiency.
- **Engineered a co-lending** platform with dynamic lender selection, developing a selector API integrated with data science models to route applications to optimal lenders based on real-time A/B/C risk scoring.
- **Designed and maintained** a robust E-NACH system, handling multi-step net banking mandate flows with reliable state management, ensuring transactional consistency, idempotency, and rollback handling for failure scenarios.
- **Developed Aadhaar-based** E-Sign authorization system (UIDAI integration), enabling secure and compliant digital signing flows, significantly improving turnaround time and reducing manual intervention.
- **Built a dynamic credit** scoring orchestration layer, intelligently selecting between providers (CIBIL, CRIF, Experian) based on partner configuration and real-time requirements, optimizing response time and hit rates.